

MBAN 5140 F - VISUAL ANALYTICS & MODELLING

PROJECT # 02 - REPORT



WARISHA SAAD

220988275



Child and Family Centre

Executive Summary

The EarlyON Child and Family Centre Dashboard provides an interactive tool for parents to find childcare programs based on location, availability, and accessibility. Programs are categorized by proximity (Closest, Mid-range, Farthest, Indigenous-focused) and availability (Available, Limited, Unknown), with 218 programs open and 19 uncertain. Most programs operate in the morning and afternoon, with limited evening options. Over 50% are community-based rather than school-based, affecting structured learning access. Language analysis reveals a 99.16% gap in Francophone support. Monday hosts the most sessions, while weekend options are scarce. A ward-level breakdown shows York South-Weston and Toronto Danforth have the most programs, while Willowdale and Don Valley North have fewer.

A. User Story

Meet Sarah – A Parent Looking for an EarlyON Program Sarah is a busy mother of a three-year-old and wants to find an EarlyON program that is:

- **Nearby** – She doesn't want to travel too far.
- **Available** – She needs a program with open spots.
- **Right Timing** – She prefers a program that fits her schedule.
- **In a School or Community Center** – She wants a structured environment.
- **Matching Language Needs** – She wants bilingual support for her child.

1. Sarah Opens the EarlyON Dashboard

Sarah visits the EarlyON Dashboard on her **laptop** to explore available programs for her child.

What she sees:

Interactive Map with various program locations.

- Program locations are **color-coded**:
 - **Green** → Closest program locations.
 - **Orange** → Mid-range program location.
 - **Red** → Farthest program location.
 - **Purple** → Indigenous-focused program location.

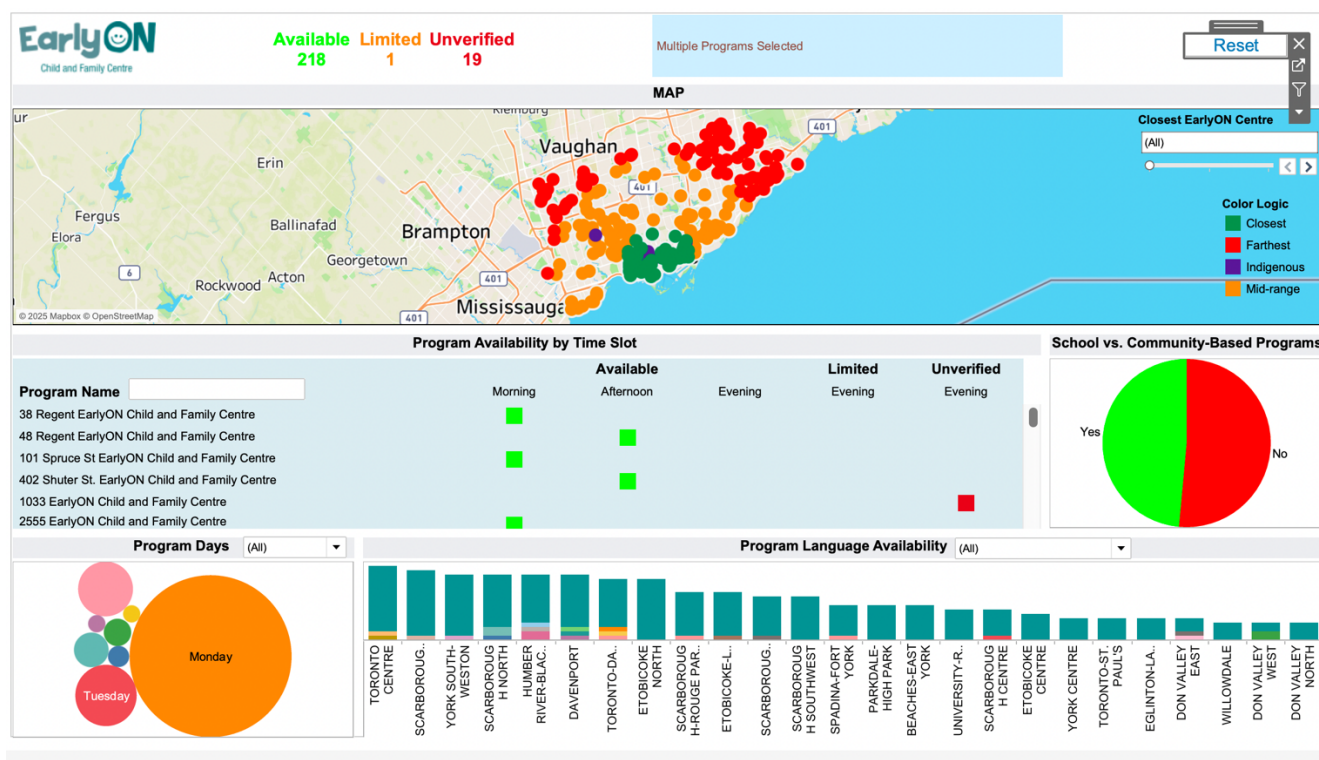


Figure # 01: EarlyOn Child and Family Center

2. Sarah Searches for Nearby Programs

Sarah doesn't want to travel far. She **zooms in on her neighborhood** and **clicks a program on the map** to see details.

What happens?

- The **list updates dynamically**, showing only programs within her **selected area**.

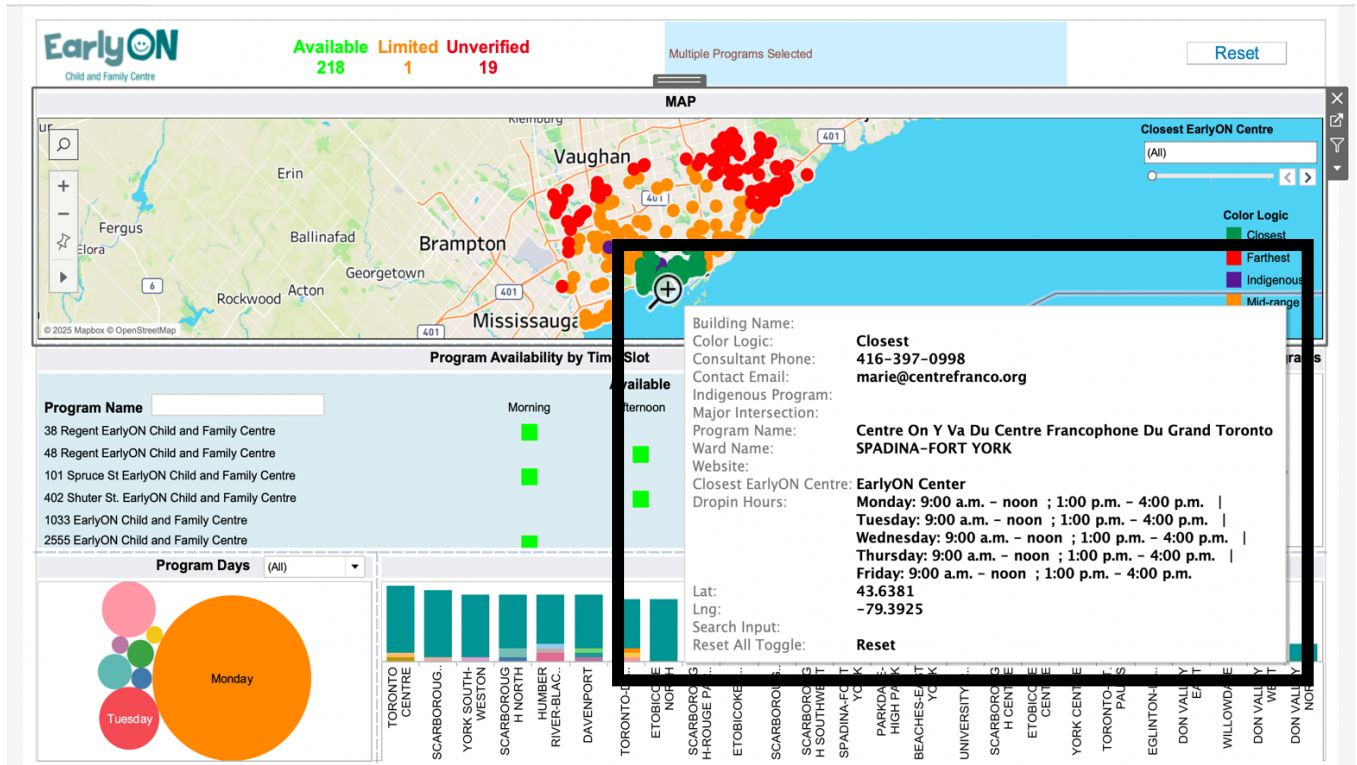


Figure # 02: Clicks on a program's location

Outcome: Sarah quickly finds programs closest to her home.

3. Sarah Checks Program Availability

Sarah only wants programs that **have open spots**, so she uses the **Program Availability** filter.

She sees:

- **Available** (Green) → Programs with open slots.
- **Limited** (Orange) → Programs with few spots left.
- **Unverified** (Red) → Availability not confirmed.

What happens?

- The map **hides unavailable programs**, showing **only programs she can join**.

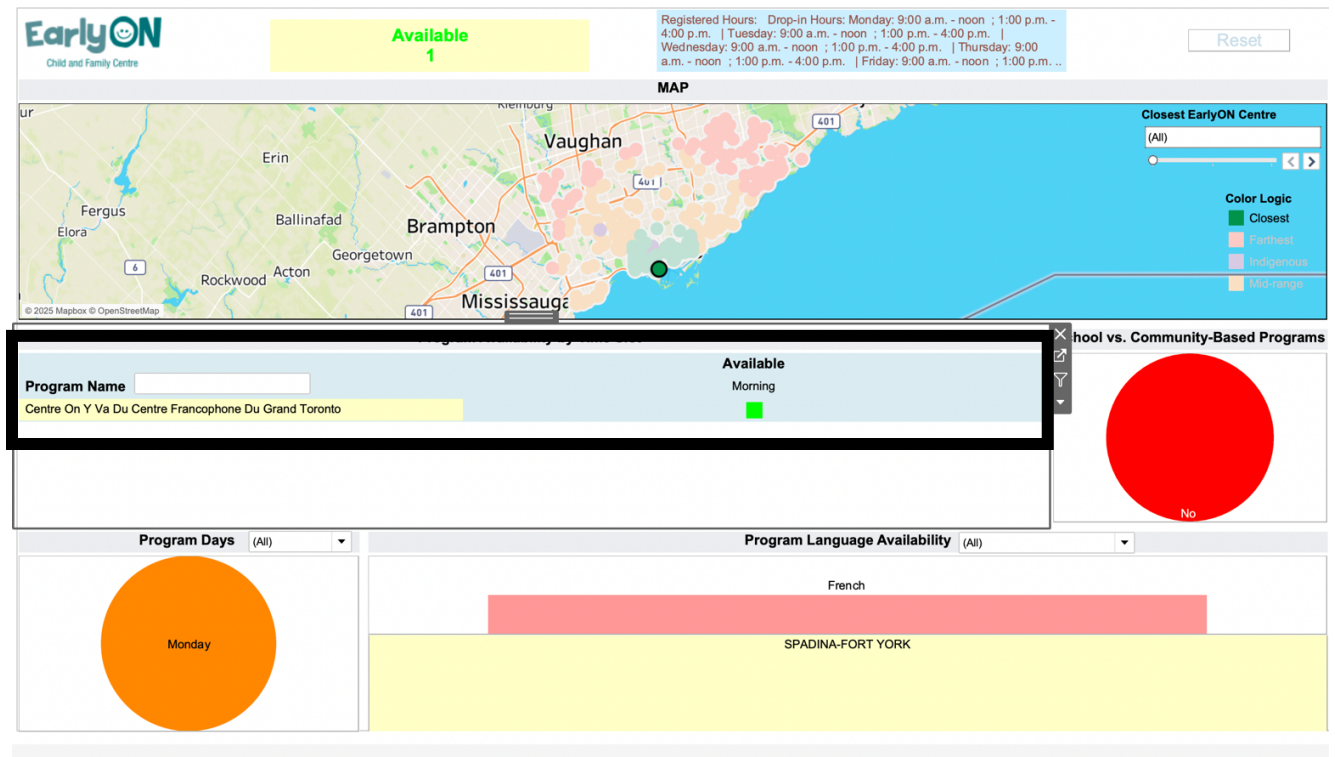


Figure # 03: Selected location

Outcome: She focuses on programs **she can enroll in today**.

4. Sarah Filters by Time (Morning/Afternoon/Evening)

Sarah prefers **morning sessions**, so she uses the **Program Day** filter to narrow down her search.

What happens?

- The **list updates** to show programs that operate in the **morning**.
- She **can also check evening options** if needed.

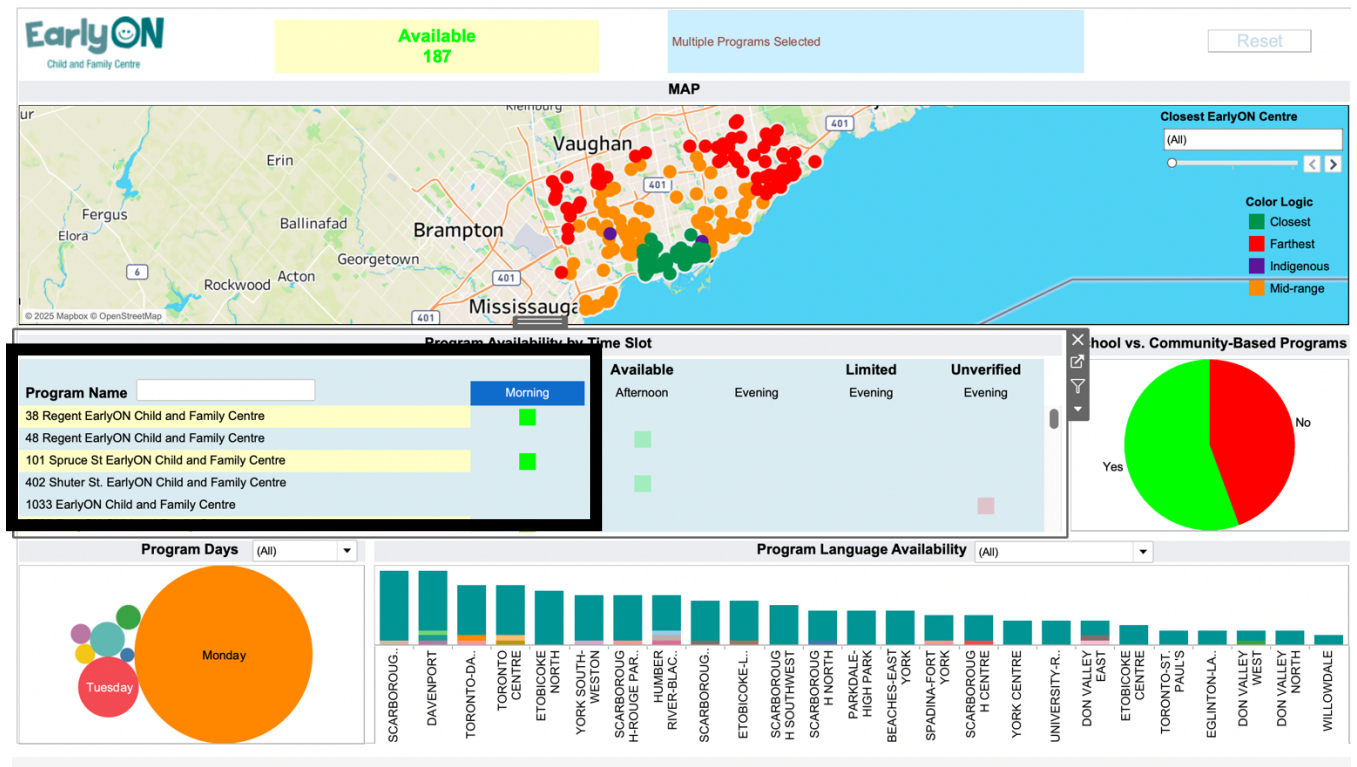


Figure # 04: Selected Morning Time Slot

Outcome: Sarah ensures the program fits her schedule.

5. Sarah Chooses Language Support

Sarah wants her child to learn in **French or Spanish**, so she uses the **Languages** filter.

What happens?

- The **dashboard updates to show only programs** that offer support in **French or Spanish**.
- Other programs are **removed from the list**.

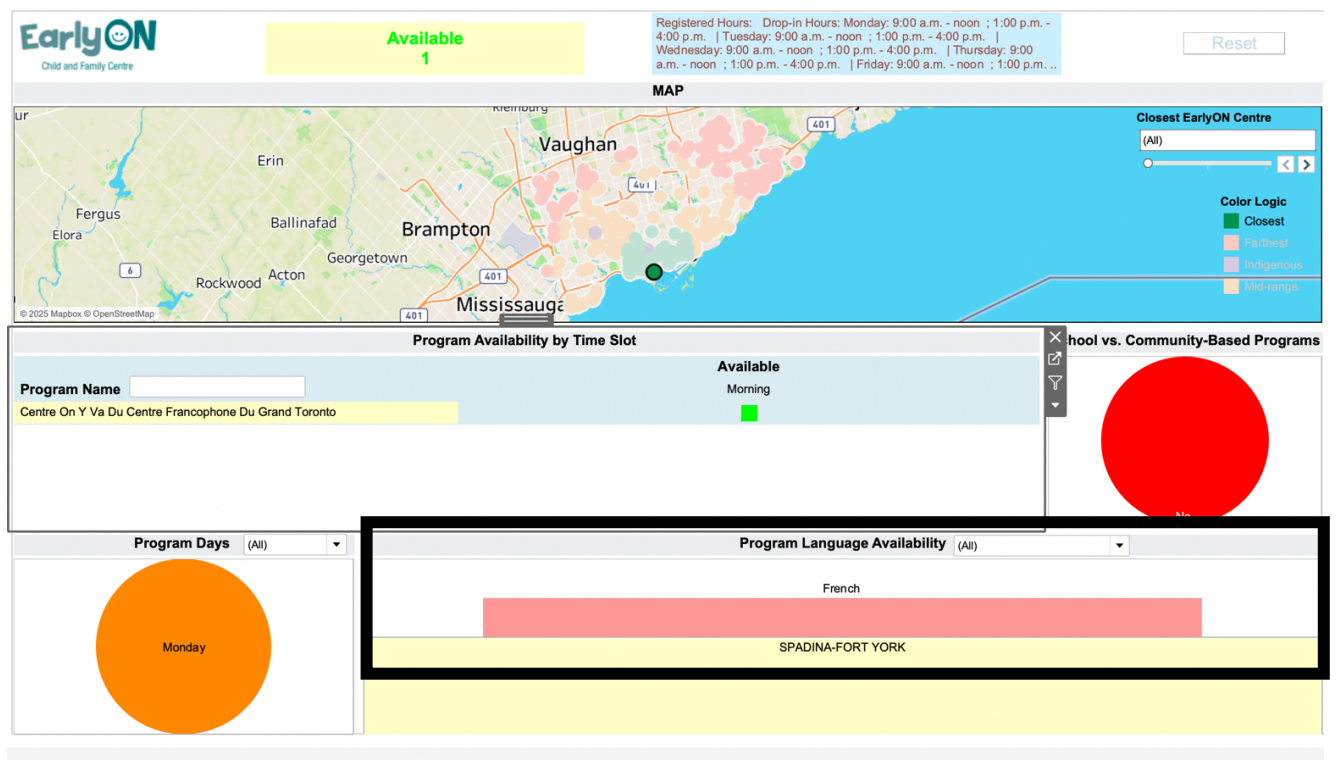


Figure # 05: Selected Language

Outcome: She finds a bilingual program for her child.

6. Sarah Chooses Between School-Based & Community Centers

Sarah prefers **structured learning** and selects **"Yes"** under **Located in School**.

What happens?

- The **dashboard updates**, showing only **school-based programs**.
- If she **deselects it**, she sees **community-based options**.

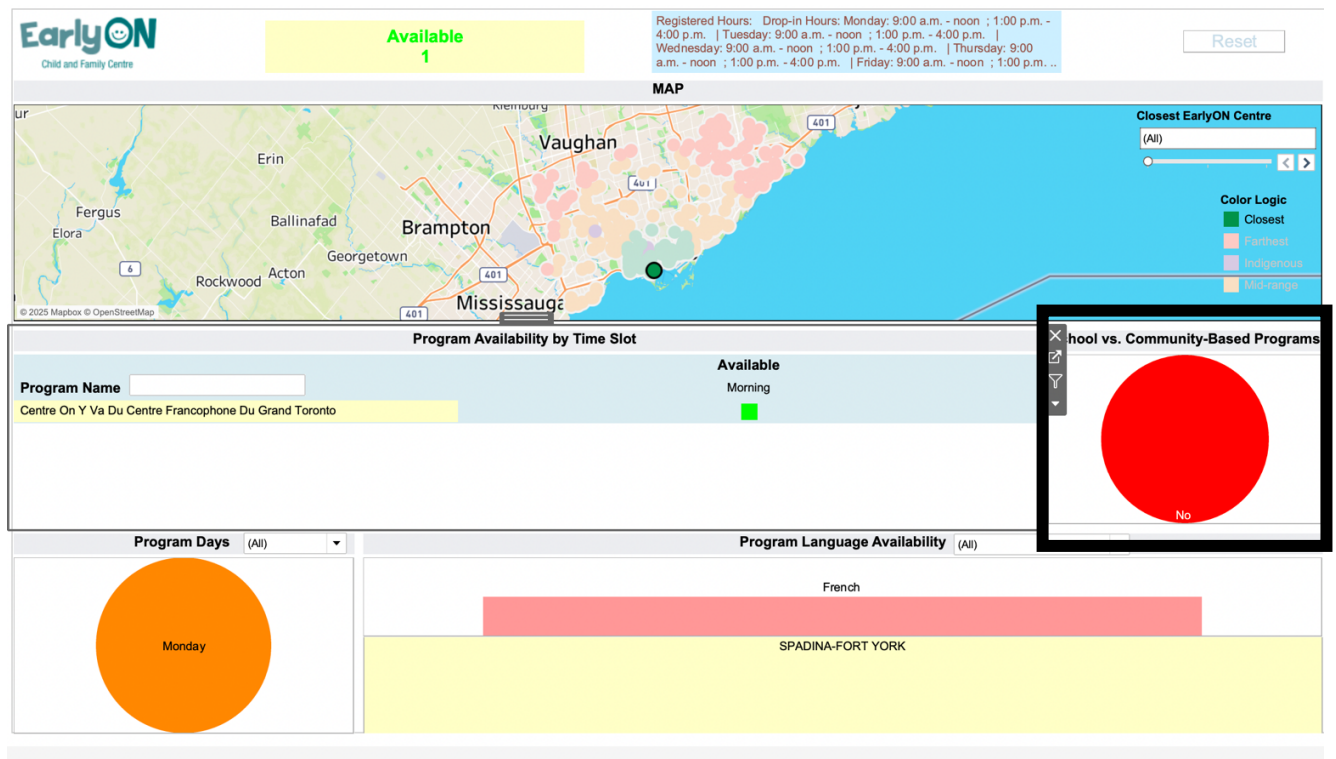


Figure # 06: School Based vs Community based Program Selection

Outcome: She chooses the best learning environment for her child.

7. Program Availability Confirmation for Selected Day

When Sarah selects a specific **program**, the **Program Days** section updates accordingly.

What happens?

- The **Program Days** visualization highlights **Monday**, indicating that the selected program operates on that day.
- Other days remain inactive, as the program is not available on them.

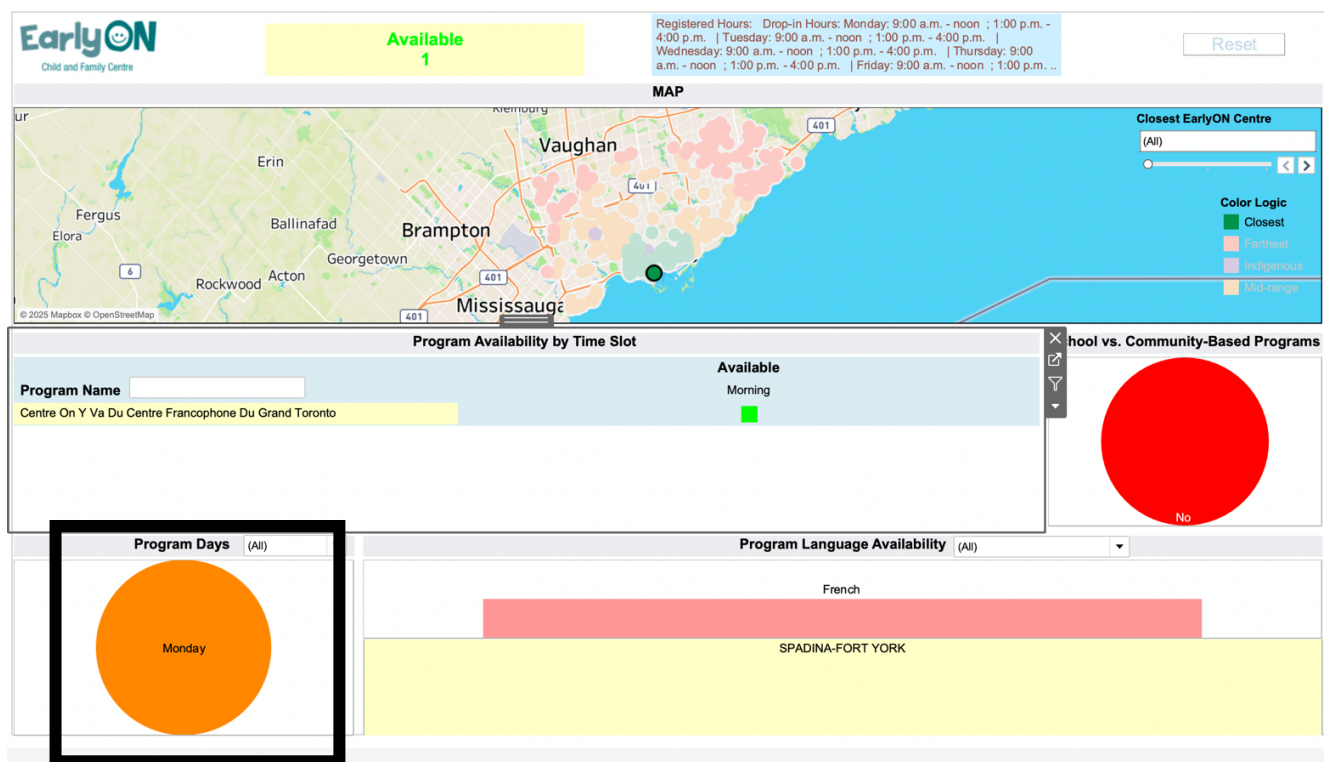


Figure # 07: Selected Language

Outcome: Sarah confirms that the selected program is available on Monday, helping her plan accordingly.

8. Sarah Clicks “Reset” to Start Over

After testing different filters, Sarah wants to **see all programs again**.

What happens?

- She **clicks the Reset button** → **All filters clear instantly.**
- The **dashboard resets to show everything.**

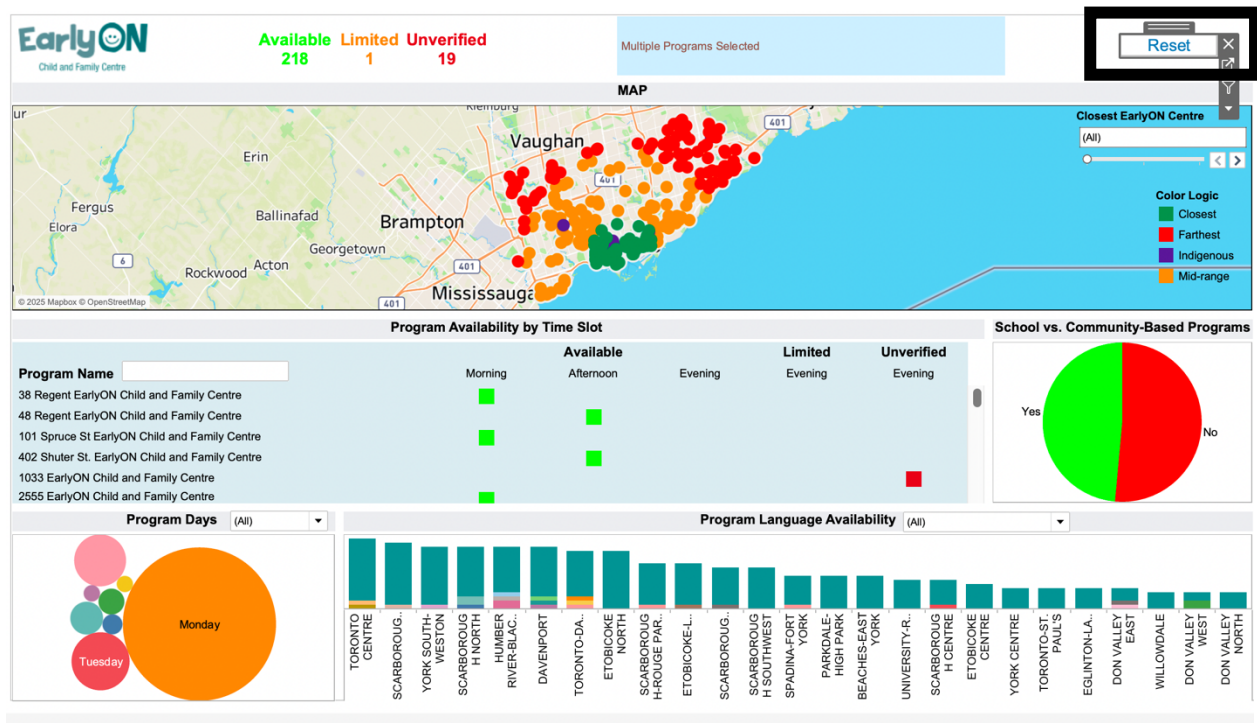


Figure # 08: Reset Button

Outcome: She **effortlessly starts a new search** without manually changing filters.

Final Step: Sarah Registers for the Best Program!

B. Insights: Data Transformations & Feature Engineering

Several transformations were applied to structure the raw data into **useful dashboard components**.

Feature Created	Data Used	Visualization Approach	Findings
Map	Distance to Reference	Categorized programs into Closest (Green), Mid-range (Orange), Farthest (Red), and Indigenous (Purple) based on distance.	- Majority of programs fall into the mid-range and farthest categories, indicating accessibility challenges for some users.
Program Availability Status	Program Availability	Mapped values to Available (Green), Limited (Orange), and Unknown (Blue) to visually indicate accessibility.	- 218 programs are Available. - 1 program is Limited. - 19 programs have Unknown availability. - Most programs have open slots, but uncertainty exists in some locations.
Time Slot Category	Drop-in, Registered, and Virtual Hours	Extracted time values and categorized them into Morning, Afternoon, and Evening.	- Most programs occur in the Morning and Afternoon, with fewer in the Evening. -Virtual programs are less frequent than in-person programs.
School vs. Community-Based Programs	Located in School	Converted binary values (Yes/No) into an interactive filter for school-based programs.	- More than 50% of programs are NOT school-based, indicating a preference for community centers or other venues. - Some areas lack school-based programs, limiting access for parents seeking structured environments.

Language Filtering	Languages	Created a multi-select filter to help parents find bilingual programs.	- Programs are available in multiple languages, but English remains dominant. - Some wards lack bilingual support, limiting access for non-English-speaking parents.
Program Day Analysis	Registered & Drop-in Data	Aggregated program data by days of the week to analyze attendance trends.	- Monday has the highest number of programs, while Tuesday and Friday have fewer offerings. - Weekend programs are limited, impacting parents who need sessions on Saturdays and Sundays.
Program Count by Ward	Ward Name & Extracted Time Range	Bar chart displaying program distribution per ward.	- York South-Weston and Toronto Danforth have the most programs. - Willowdale & Don Valley North have significantly fewer programs.
French Language Program Analysis	French Language Program	Pie chart analyzing the proportion of French-language programs.	- 99.16% of programs are NOT French-language-based. - Only 0.84% of programs cater to Francophone families, revealing a gap in French-language support.
Program Type Analysis	Drop-in, Registered, and Virtual Hours	Bar chart visualizing the count of Drop-in, Registered, and Virtual programs.	- Drop-in programs are the most common (140+ programs). - Registered programs rank second (~60) - Virtual programs are very limited making remote access difficult.

Table # 01: Insights

C. Rationale: EarlyON Child and Family Centre

Topic	Reason for Selection
Community Services	EarlyON centers provide essential childcare, education, and community engagement services for families.
Locations and Mapping	The dashboard visually maps EarlyON centers, helping parents find nearby programs based on proximity and availability.
City Government	EarlyON programs are publicly funded, and insights from the dashboard can support policy decisions on accessibility and expansion.
Mobility (Transportation)	Distance-based categorization (Closest, Mid-range, Farthest) helps parents evaluate travel feasibility and accessibility.
Poverty Reduction & Affordable Housing	EarlyON services are free, benefiting low-income families. Identifying service gaps ensures equitable access.
Health	The centers offer parenting resources, early childhood education, and culturally inclusive language support, improving child and family well-being.
Fiscal Responsibility	Data insights help allocate funding efficiently, ensuring resources are distributed based on program usage and demand.
Development and Infrastructure	Identifying service gaps can inform city planning for expanding EarlyON centers in underserved areas.
Climate Change (Indirect Impact)	Improved access to local programs can reduce long commutes, supporting sustainable urban mobility and lowering emissions.

Table # 02: Rationale

References:

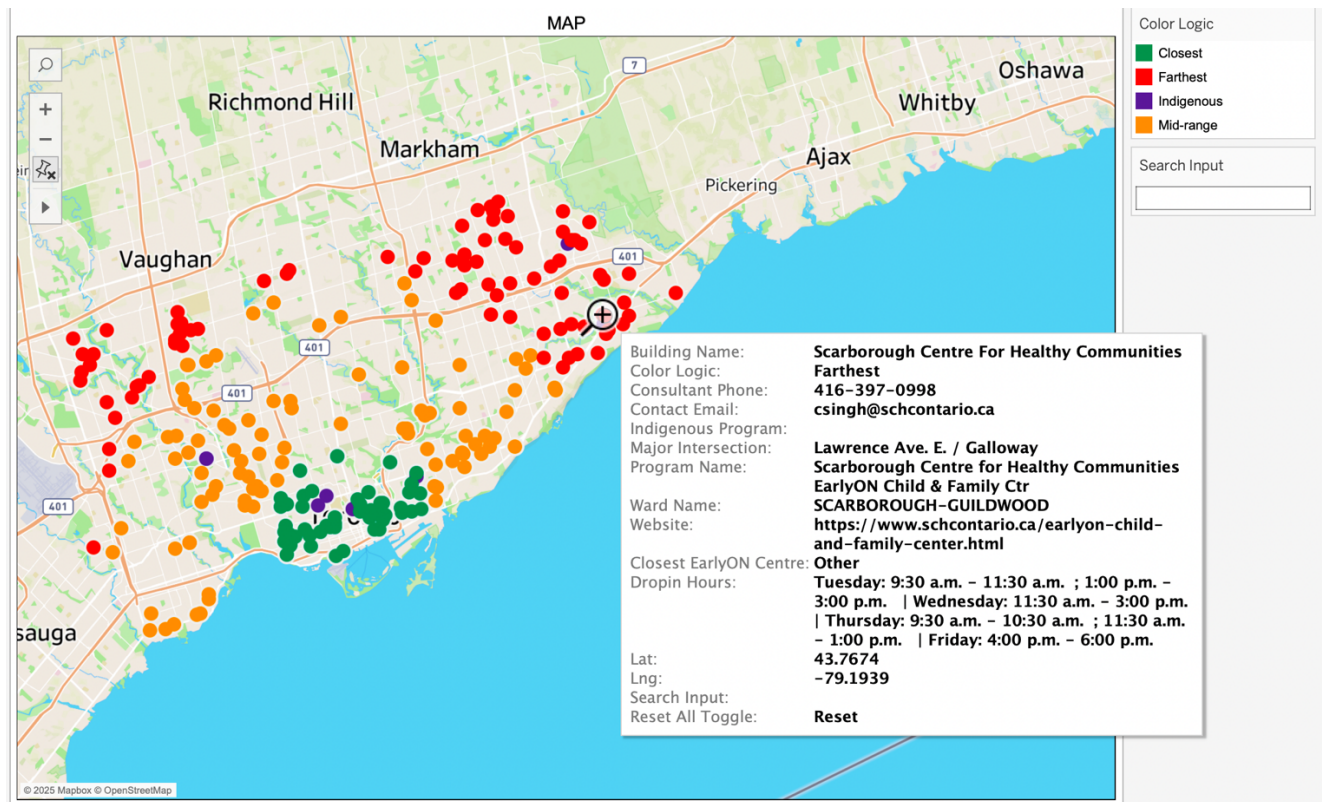
City of Toronto. *Child & Family Programs*. City of Toronto, n.d.
<https://www.toronto.ca/community-people/children-parenting/children-programs-activities/child-family-programs-and-groups/child-family-programs/>.

City of Toronto. *EarlyON Child and Family Centres*. City of Toronto Open Data, n.d.
Web. 4 Mar. 2025. <https://open.toronto.ca/dataset/earlyon-child-and-family-centres/>.

Appendices:

1. EarlyON Program Location Map

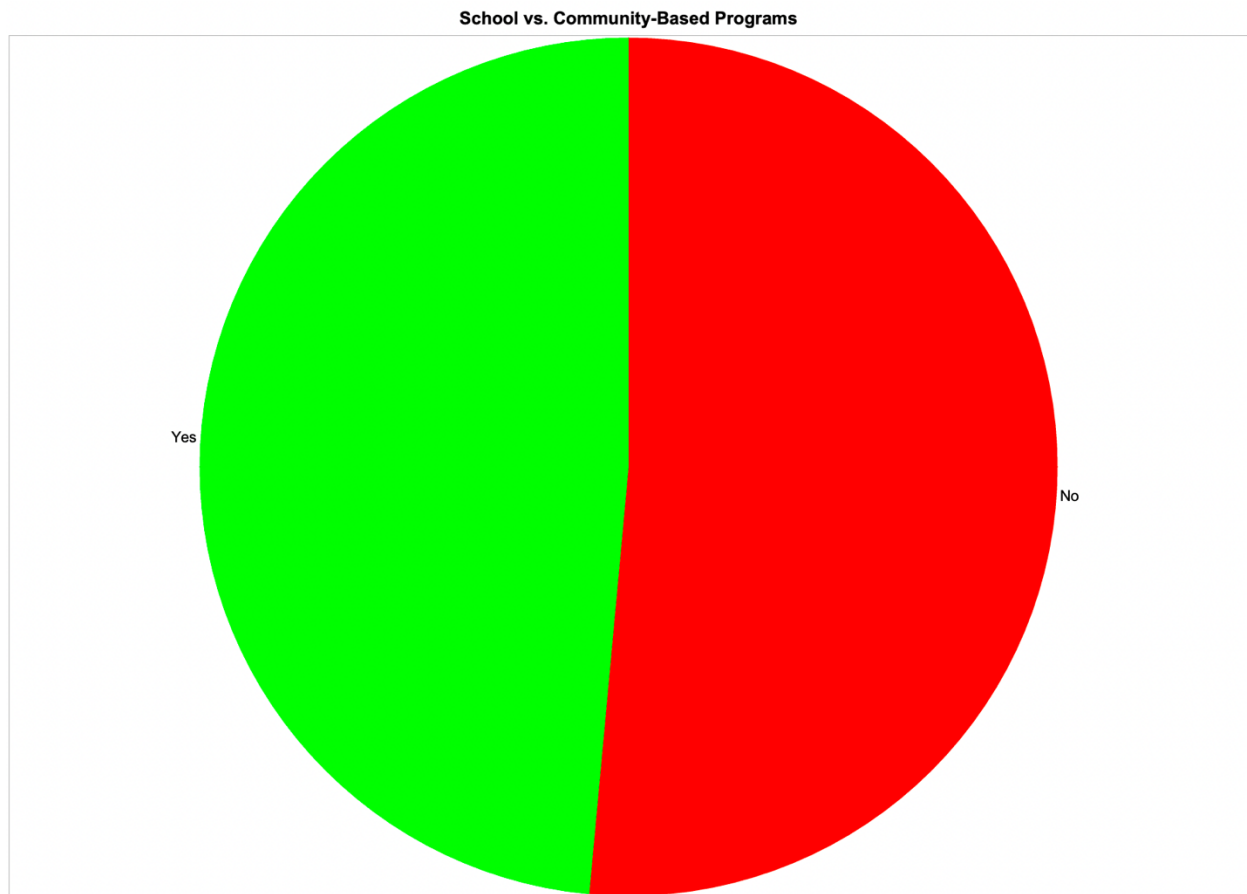
This map provides a geographic representation of EarlyON Child and Family Centres across the region, categorized by distance and program focus. Each program location is plotted based on its latitude and longitude, and color-coded to indicate proximity and type.



Visualization # 01: EarlyON Program Location Map

2. School vs. Community-Based Programs

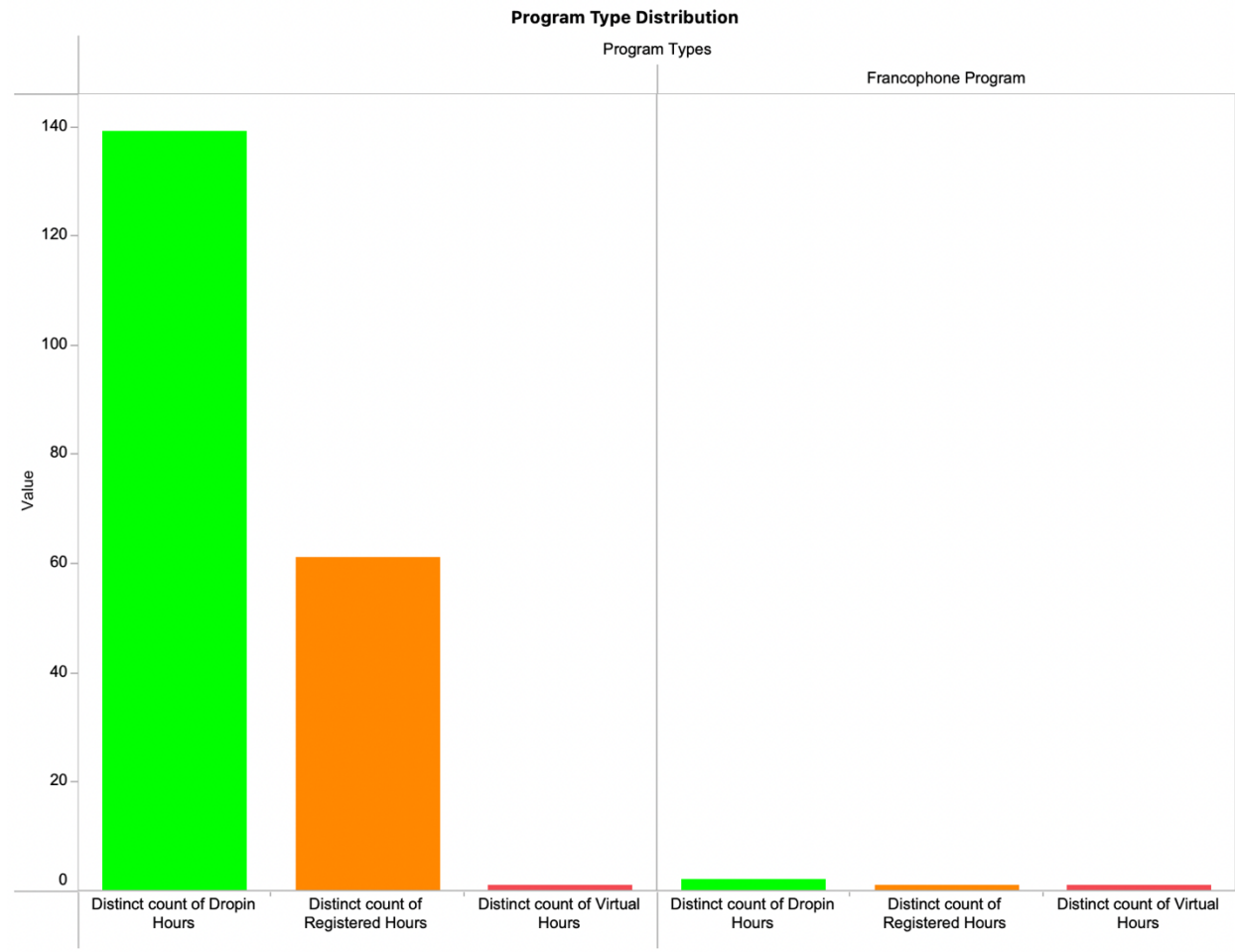
This bar chart visualizes the distribution of EarlyON programs based on whether they are located within a school setting or in other community-based locations.



Visualization # 02: School vs. Community-Based Programs

3. Program Type Distribution

This bar chart visualizes the distribution of EarlyON Child and Family Centre programs by type, highlighting **Drop-in**, **Registered**, and **Virtual programs**. Each type is color-coded for clear differentiation.



Visualization # 03: Program Type Distribution

4. Program Availability by Time Slot

This visualization presents an overview of **EarlyON Child and Family Centre programs** along with their **availability status and time slots**. The chart helps parents identify **which programs operate in the morning, afternoon, or evening** while also displaying their availability status.

Program Availability by Time Slot					
Program Name	Morning	Available Afternoon	Evening	Limited Evening	Unverified Evening
38 Regent EarlyON Child and Family Centre	■				
48 Regent EarlyON Child and Family Centre		■			
101 Spruce St EarlyON Child and Family Centre	■				
402 Shuter St. EarlyON Child and Family Centre		■			
1033 EarlyON Child and Family Centre					■
2555 EarlyON Child and Family Centre	■				
2700 Dufferin EarlyON Child and Family Centre	■				
4139 Sheppard Ave East EarlyON Child and Family Centre at ACSA	■				
Abiona EarlyON Child and Family Centre	■				
Access Alliance EarlyON Child and Family Centre		■			
ACSA EarlyON Child and Family Centre at Goldhawk Recreation Room			■		
Agincourt Community Services EarlyON Child and Family Centre			■		
Agnes Mac Phail P.S EarlyON Child and Family Centre		■			
Albion Library Satellite EarlyON Child and Family Centre		■			
Alexander Muir/Gladstone EarlyON Child and Family Centre	■				
Alexandra Park EarlyON Child and Family Centre	■				
Amesbury Community Centre EarlyON					■
Annette Public Library EarlyON Child and Family Centre	■				
Anson Taylor EarlyON Child and Family Centre		■			
Applegrove Connection EarlyON Child and Family Centre	■				
Applegrove EarlyON Child and Family Centre	■				
Banting Best EarlyON Child and Family Centre	■				
Barbara Frum Library EarlyON					■
Bathurst-Finch Hub EarlyON Child and Family Centre	■				
Birchmount Bluffs EarlyON Child and Family Centre	■				
Black Creek Library					■
Braeburn EarlyON Child and Family Centre			■		
Braeburn Tandridge EarlyON Child and Family Centre	■				
Bridlewood EarlyON Centre	■				

Visualization # 04: Program Availability by Time Slot

5. Program Count by Ward and Extracted Time Range

This visualization displays the **distribution of program time slots across different city wards**, measured by the **count of extracted time ranges**. The analysis helps to understand **which areas have more program availability and which are underserved**.

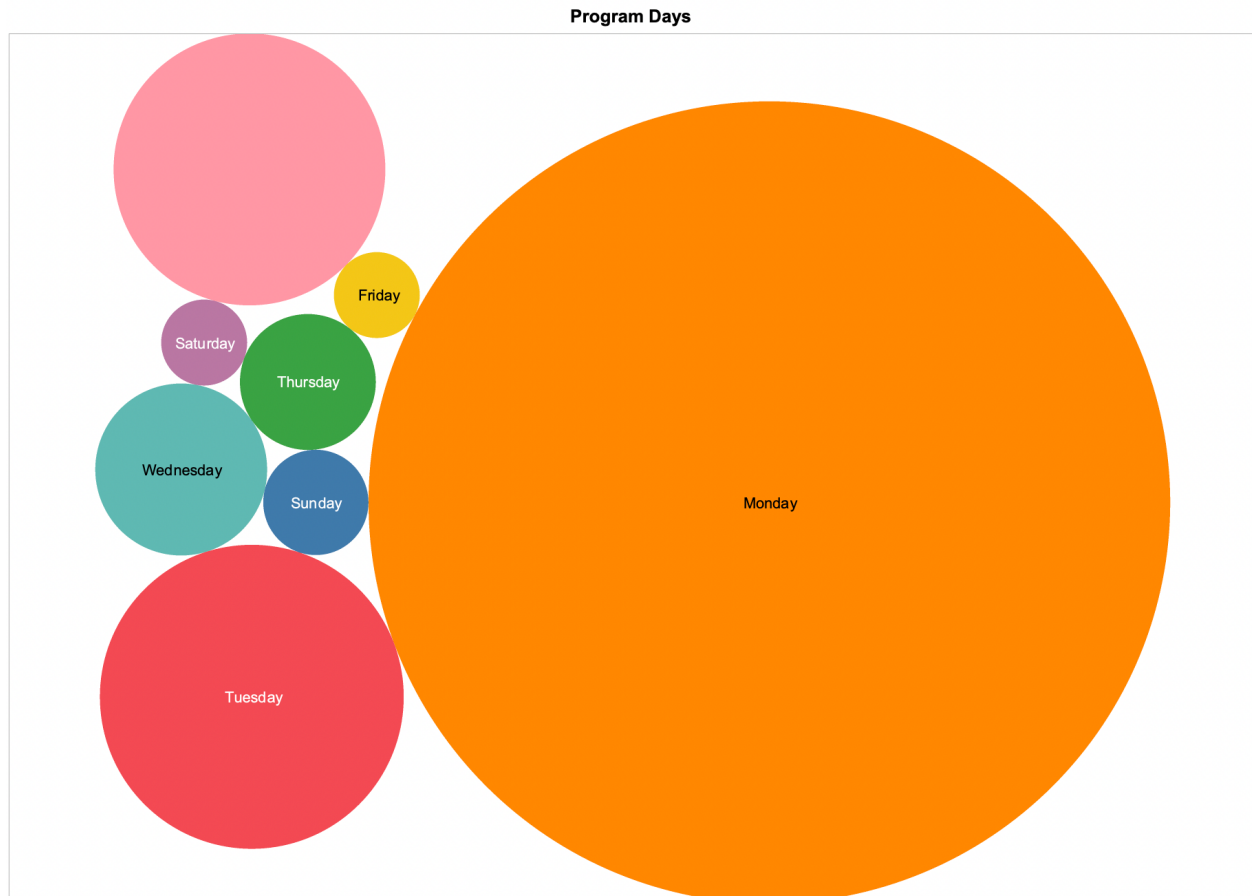
Program Count by Ward and Time Slot Analysis

Ward Name	
BEACHES-EAST YORK	8
DAVENPORT	15
DON VALLEY EAST	5
DON VALLEY NORTH	4
DON VALLEY WEST	4
EGLINTON-LAWRENCE	5
ETOBICOKE CENTRE	6
ETOBICOKE NORTH	14
ETOBICOKE-LAKESHORE	11
HUMBER RIVER-BLACK CREEK	15
PARKDALE-HIGH PARK	8
SCARBOROUGH CENTRE	7
SCARBOROUGH NORTH	15
SCARBOROUGH SOUTHWEST	10
SCARBOROUGH-AGINCOURT	10
SCARBOROUGH-GUILDWOOD	16
SCARBOROUGH-ROUGE POINT	11
SPADINA-FORT YORK	8
TORONTO CENTRE	17
TORONTO-DANFORTH	14
TORONTO-ST. PAUL'S	5
UNIVERSITY-ROSEDALE	7
WILLOWDALE	4
YORK CENTRE	5
YORK SOUTH-WESTON	15

Visualization # 05: Program Count by Ward and Extracted Time Range

6. Program Days Analysis

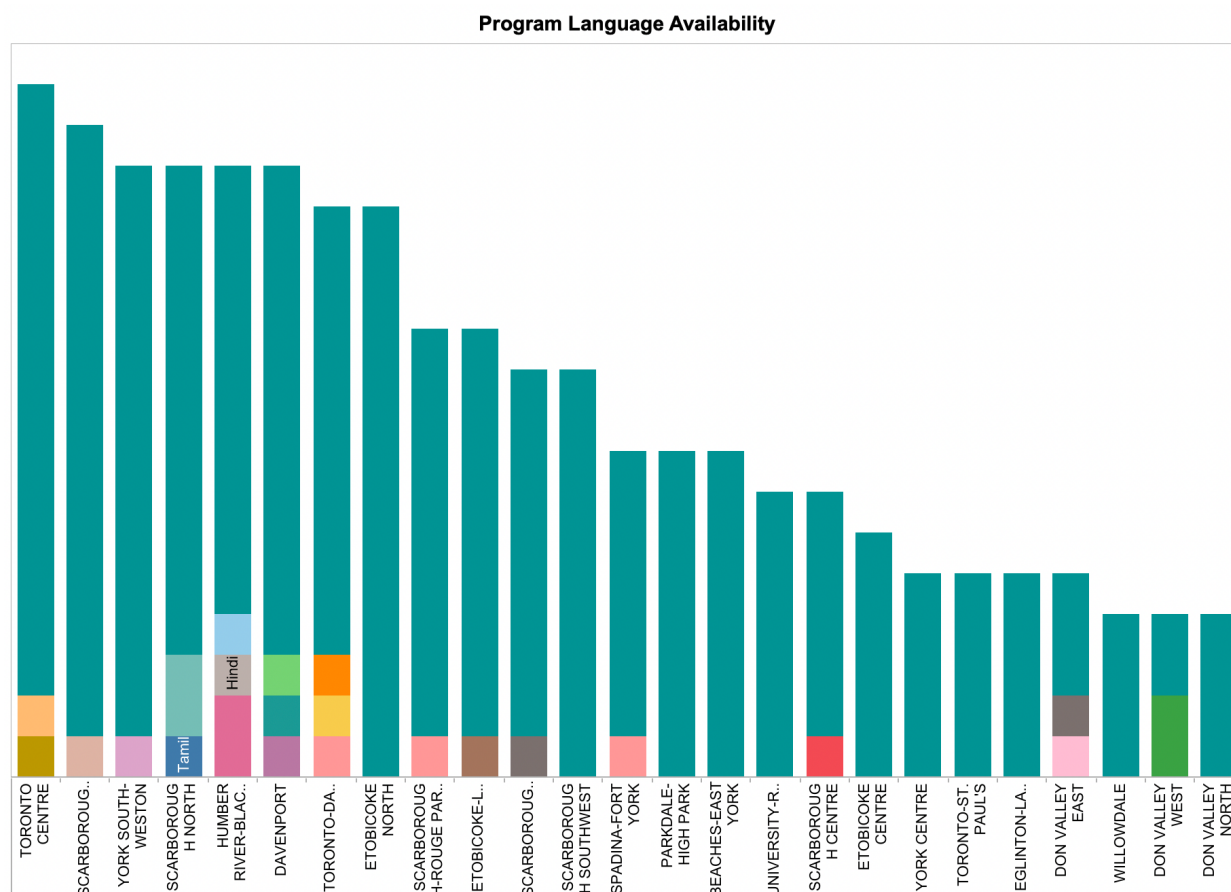
This visualization represents the distribution of EarlyON programs across different days of the week. The bar chart illustrates the frequency of programs scheduled each day, helping identify trends in program availability.



Visualization # 06: Program Days Analysis

7. Program Language Availability

The bar chart titled "Languages" provides insights into the language distribution across different wards for EarlyON programs. The majority of programs in each ward are conducted in English, as indicated by the dominant, teal-colored bars. However, some wards have programs offering additional language support, represented by smaller sections of different colors at the bottom of the bars.

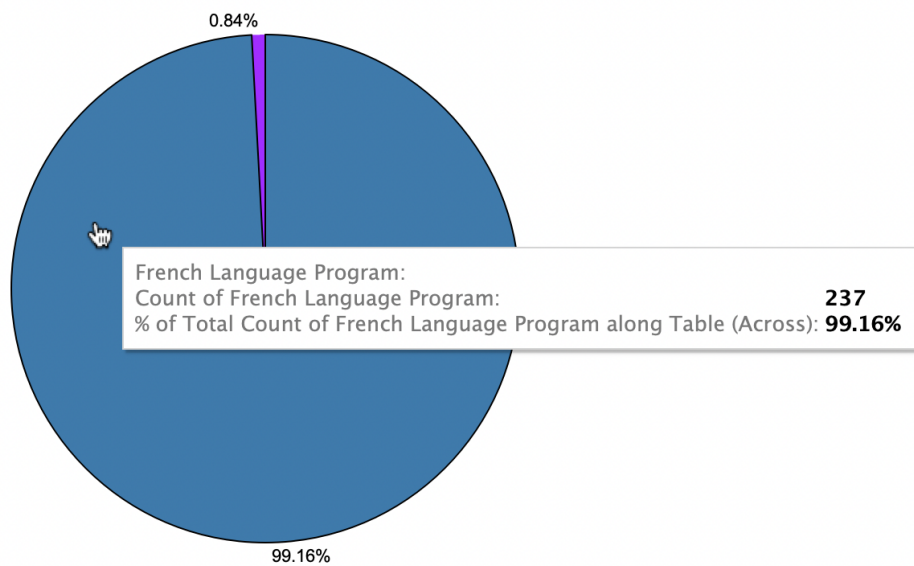


Visualization # 07: Program Language Availability

8. Availability of French-Language Programs

This pie chart represents the availability of French-language programs in EarlyON Child and Family Centres. The visualization indicates that **only 0.84%** of the programs offer French-language support, while **99.16%** of the programs do not provide French-language services. This highlights a significant gap in accessibility for Francophone families seeking EarlyON services. The findings suggest a need for increased bilingual or French-language programs to better support diverse communities.

Availability of French-Language Programs



Visualization # 08: Availability of French-Language Programs